



High Level Problems (HLP)

Marked Questions can be used as Revision Questions.

PART - I : DIFFERENTIATION

Find the derivative of functions using quotient rule.

1. $g(x) = \frac{x^2 - 4}{x + 0.5}$

2. Suppose u and v are differentiable functions of x and that
 $u(1) = 2$, $u'(1) = 0$ $v(1) = 5$ $v'(1) = -1$.
 Find the values of the following derivatives at $x = 1$.

(a) $\frac{d}{dx}(uv)$ (b) $\frac{d}{dx}\left(\frac{u}{v}\right)$ (c) $\frac{d}{dx}\left(\frac{v}{u}\right)$ (d) $\frac{d}{dx}(7v - 2u)$.

Find $\frac{ds}{dt}$.

3. $s = \frac{1 + \operatorname{cosec} t}{1 - \operatorname{cosec} t}$

4. $s = \frac{\sin t}{1 - \cos t}$

Find $\frac{dy}{dx}$ as a function of x .

5. $y = \sin^3 x$

6. $y = 5 \cos^{-4} x$.

Find the derivatives of the functions

7. $r = (\operatorname{cosec} \theta + \cot \theta)^{-1}$

8. $r = -(\sec \theta + \tan \theta)^{-1}$

PART - II : INTEGRATION

Find an antiderivative for each function. Do as many as you can mentally. Check your answer by differentiation.

1. (a) $\csc x \cot x$ (b) $-\csc 5x \cot 5x$ (c) $-\pi \csc \frac{\pi x}{2} \cot \frac{\pi x}{2}$

2. $(1 + 2 \cos x)^2$

Evaluating Integrals.

Check your answers by differentiation.

3. $\int \left(\frac{\sqrt{x}}{2} + \frac{2}{\sqrt{x}} \right) dx$

4. $\int \left(8y - \frac{2}{y^{1/4}} \right) dy$

5. $\int 2x(1 - x^{-3}) dx$

6. $\int (-3 \csc^2 x) dx$

7. $\int \left(-\frac{\sec^2 x}{3} \right) dx$

8. $\int \frac{\csc \theta \cot \theta}{2} d\theta$

9. $\int \frac{2}{5} \sec \theta \tan \theta d\theta$

10. $\int (4 \sec x \tan x - 2 \sec^2 x) dx$

11. $\int \frac{1}{2} (\csc^2 x - \csc x \cot x) dx$

12. $\int (\sin 2x - \csc^2 x) dx$

13. $\int (2 \cos 2x - 3 \sin 3x) dx$

14. $\int 4 \sin^2 y dy$

15. $\int \frac{\csc \theta}{\csc \theta - \sin \theta} d\theta$





Evaluate Integrals by substitution method.

16. $\int \frac{1}{\sqrt{5s+4}} ds$ 17. $\int 3y\sqrt{7-3y^2} dy$ 18. $\int \sin^5 \frac{x}{3} \cos \frac{x}{3} dx.$
19. $\int \tan^7 \frac{x}{2} \sec^2 \frac{x}{2} dx$ 20. $\int r^2 \left(\frac{r^3}{18} - 1 \right)^5 dr$ 21. $\int r^4 \left(7 - \frac{r^5}{10} \right)^3 dr$
22. $\int x^{1/3} \sin(x^{4/3} - 8) dx.$ 23. $\int \csc \left(\frac{v-\pi}{2} \right) \cot \left(\frac{v-\pi}{2} \right) dv$
24. $\int \sqrt{\cot y} \csc^2 y dy$ 25. $\int \frac{\sec z \tan z}{\sqrt{\sec z}} dz$ 26. $\int \frac{1}{t^2} \cos \left(\frac{1}{t} - 1 \right) dt$

Find the definite integrals of following Functions

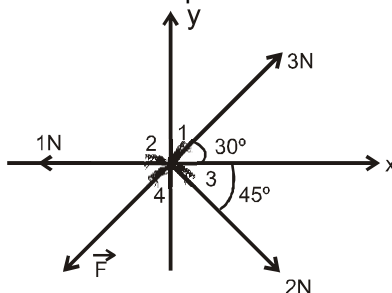
27. $\int_{1/2}^{3/2} (-2x+4) dx$ 28. $\int_{-2}^1 |x| dx$

Evaluate definite integrals of following functions

29. $\int_0^{\pi/2} \theta^2 d\theta$ 30. $\int_0^{3b} x^2 dx$

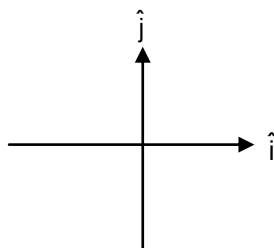
PART - III : VECTOR

- Four forces of magnitudes P, 2P, 3P and 4P act along the four sides of a square ABCD in cyclic order. Use the vector method to find the resultant force.
- A sail boat sails 2km due East, 5km 37° South of East and finally has an unknown displacement. If the final displacement of the boat from the starting point is 6km due East, the third displacement is _____.
- The resultant of two vectors $\mathbf{u}$ and $\mathbf{v}$ is perpendicular to the vector $\mathbf{u}$ and its magnitude is equal to half of the magnitude of vector $\mathbf{v}$. Find the angle between $\mathbf{u}$ and $\mathbf{v}$.
- Let the resultant of three forces of magnitude 5N, 12N & 13N acting on a body be zero. If $\sin 23^\circ = (5/13)$, find the angle between the 5N force & 13N force.
- Two vectors $\vec{A}$ & $\vec{B}$ have the same magnitude. Under what circumstances does the vector $\vec{A} + \vec{B}$ have the same magnitude as $|\vec{A}|$ or $|\vec{B}|$. When does the vector difference $\vec{A} - \vec{B}$ have this magnitude?
- The resultant of $\vec{P}$ and $\vec{Q}$ is $\vec{R}$. If magnitude of $\vec{Q}$ is doubled, magnitude of resultant is also doubled, when direction of $\vec{Q}$ is reversed from initial condition then magnitude of resultant is again doubled, find $P : Q : R$.
- If five consecutive sides of a regular hexagon represent five unit vectors acting in the same sense, find their resultant vector. (Taking first side on x-axis)
- Four ants 1, 2, 3 and 4 are pulling a grain with force of magnitudes 3 N, 1N, 2N and $|\vec{F}|$ N as shown in the Fig. Find force $\vec{F}$ if the grain remains in equilibrium under the action of the above forces.

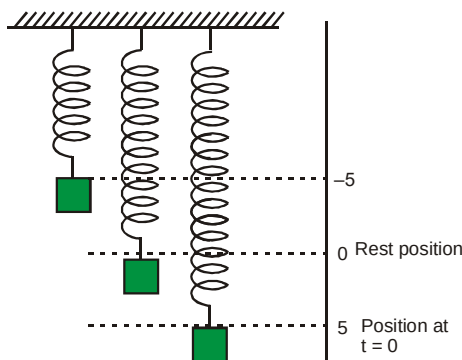




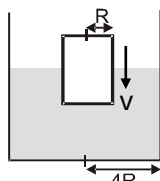
9. An insect moves in a circular path of radius R . If it rotates through an angle θ , find its displacement means $\vec{S} = \Delta \vec{r}$, where $\Delta \vec{r}$ is "the change in position vector".



10. Find the vector equation of a line which is parallel to a given vector $\vec{A}$ and passes through a given point P having position vector $\vec{r}_0$.
11. By using the concept of scalar product prove that $|\vec{A} + \vec{B}| = \sqrt{|\vec{A}|^2 + |\vec{B}|^2 + 2|\vec{A}||\vec{B}|\cos\theta}$
12. The vector $\vec{A}$ varies with time as $\vec{A} = t\hat{i} - \sin\pi t\hat{j} + t^2\hat{k}$. Find the derivative of the vector at $t = 1$.
13. A body hanging from a spring (fig.) is stretched 5 units beyond its rest position and released at time $t = 0$ to oscillate up and down. Its position at any later time t is $s = 5 \cos t$. What are its velocity and acceleration at time t ?



14. A cylinder of radius R is moving down with speed v into water placed in a cylinder of radius $4R$ as shown. Find the rate by which



(a) Water surface is rising

(b) Wet surface area of cylinder of radius R is increasing.

15. Find the maximum area of the rectangle that can be inscribed in a circle of radius r ?

HLP Answers

PART - I

1. $g'(x) = \frac{x^2 + x + 4}{(x + 0.5)^2}$ 2. (a) -2 (b) $\frac{2}{25}$ (c) $-\frac{1}{2}$ (d) -7
3. $\frac{-2 \operatorname{cosec} t \cot t}{(1 - \operatorname{cosec} t)^2}$ 4. $\frac{ds}{dt} = \frac{1}{\cos t - 1}$ 5. $3 \sin^2 x (\cos x)$
6. $\frac{dy}{dx} = 20 \sin x \cos^{-5} x$ 7. $\frac{\operatorname{cosec} \theta}{\cot \theta + \operatorname{cosec} \theta}$ 8. $\frac{\sec \theta}{\sec \theta + \tan \theta}$





PART - II

1. (a) $-\csc x + C$ (b) $\frac{1}{5} \csc(5x) + C$ (c) $2 \csc\left(\frac{\pi x}{2}\right) + C$
2. $3x + \sin 2x + 4 \sin x + C$
3. $\frac{x^{\frac{3}{2}}}{3} + 4x^{1/2} + C$
4. $4y^2 - \frac{8}{3}y^{3/4} + C$
5. $x^2 + \frac{2}{x} + C$
6. $3 \cot x + C$
7. $\frac{-\tan x}{3} + C$
8. $-\frac{1}{2} \csc \theta + C$
9. $\frac{2}{5} \sec \theta + C$
10. $4 \sec x - 2 \tan x + C$
11. $-\frac{1}{2} \cot x + \frac{1}{2} \csc x + C$
12. $-\frac{1}{2} \cos 2x + \cot x + C$
13. $\sin 2x + \cos 3x + C$
14. $2y - \sin 2y + C$
15. $\tan \theta + C$
16. $\frac{2}{5} (5s + 4)^{1/2} + C$
17. $-\frac{1}{3} (7 - 3y^2)^{3/2} + C$
18. $\frac{1}{2} \sin^6\left(\frac{x}{3}\right) + C$
19. $\frac{1}{4} \tan^8 \frac{x}{2} + C$
20. $\left(\frac{r^3}{18} - 1\right)^6 + C$
21. $-\frac{1}{2} \left(7 - \frac{r^5}{10}\right)^4 + C$
22. $-\frac{3}{4} \cos(x^{4/3} - 8) + C$
23. $-2 \csc\left(\frac{v - \pi}{2}\right) + C$
24. $-\frac{2}{3} (\cot^3 y)^{1/2} + C$
25. $2\sqrt{\sec z} + C$
26. $-\sin\left(\frac{1}{t} - 1\right) + C$
27. Area = 2 square units
28. Area = 2.5 square units
29. $\frac{\pi^3}{24}$
30. $9b^3$

PART - III

1. $2\sqrt{2} P$
2. 3 km in North
3. 150°
4. 113°
5. When the angle between **A** & **B** is 120° ; when it is 60°
6. $P : Q : R = \sqrt{2} : \sqrt{3} : \sqrt{2}$
7. $\vec{A}_3 = \frac{1}{2}(-\hat{i} + \sqrt{3}\hat{j})$
8. $F_4 = \sqrt{x^2 + y^2} = \sqrt{\left(\frac{3\sqrt{3}}{2} + \sqrt{2} - 1\right)^2 + \left(\frac{3}{2} - \sqrt{2}\right)^2}$ N, $\phi = \tan^{-1}\left[\frac{3 - 2\sqrt{2}}{3\sqrt{3} + 2\sqrt{2} - 2}\right]$
9. The magnitude of displacement = $|\Delta \vec{r}| = 2R \sin \frac{\theta}{2}$. The direction of $\Delta \vec{r}$ is given as $\beta = \frac{\pi}{2} + \frac{\theta}{2}$
10. $\vec{r} = \vec{r}_0 + n\hat{a}$
11. $\vec{A} + \vec{B} = \vec{R}$, say,
Since $|\vec{R}|^2 = \vec{R} \cdot \vec{R} = (\vec{A} + \vec{B}) \cdot (\vec{A} + \vec{B})$
Expanding the product, we have $|\vec{R}|^2 = \vec{A} \cdot \vec{A} + \vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{A} + \vec{B} \cdot \vec{B}$
Substituting $\vec{A} \cdot \vec{A} = A^2$, $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ and $\vec{B} \cdot \vec{B} = B^2$, we have $|\vec{R}|^2 = A^2 + B^2 + 2\vec{A} \cdot \vec{B}$
Hence $|\vec{R}| = |\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB \cos \theta}$
12. $\hat{i} + \pi \hat{j} + 2\hat{k}$
13. Velocity: $-5 \sin t$, Acceleration: $-5 \cos t$
14. (a) $\frac{dh}{dt} = \frac{v}{15}$ (b) $\frac{32\pi Rv}{15}$
15. $A_{\max} = 2r^2$